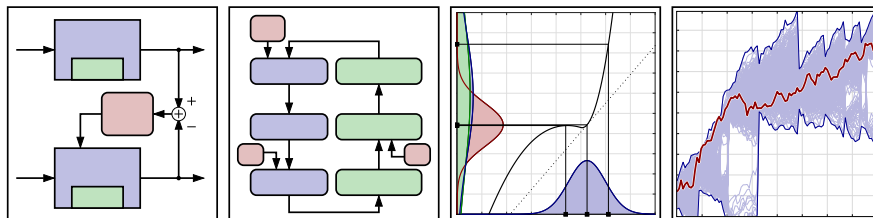


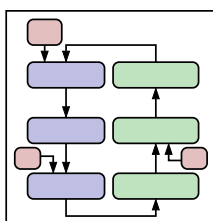


Welcome to the course!

- Welcome to **Linear Kalman Filter Deep Dive!**
- This is the second course out of four in a specialization that investigates the design and application of Kalman filters.
- This course focuses on developing a deep understanding of how the KF is derived, how to make a KF more robust, and how to modify the KF steps to accommodate scenarios that violate the standard assumptions (how to make KF actually work).



What topics will we study in this course?

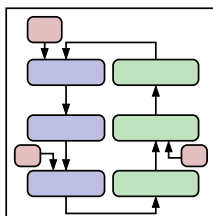


In this course, you will learn:

- Preparatory definitions, math concepts.
- How to derive the steps of the linear Kalman filter.
- How to understand the internal variables of the KF.
- How to ensure that the KF is numerically robust.
- How to modify the KF steps for scenarios that violate the standard assumptions.
- How to apply KF to a target-tracking application.



What skills will you gain in this course?



After completing the course, you'll be able to:

- Explain the purpose of each sequential-probabilistic-inference step.
- Describe the role of each internal variable of the KF.
- Modify the standard KF's steps to make it more robust numerically.
- Modify the KF's steps to enable applications that violate the standard assumptions.
- Implement a KF method for a target-tracking application.



Prerequisites for this course

Specialization prerequisites:

- A B.S. degree in Electrical, Computer, or Mechanical Engr., or
- A B.S. degree with undergraduate-level competency in the following areas:
 - **Math:** Differential and integral calculus, systems of linear equations, mechanics of linear-algebra (vector and matrix operations), basic differential equations.
 - **Probability:** Random variables (incl. pdfs, cdfs, expectations), Gaussian RVs.
 - **Programming:** MATLAB, Octave, or similar scientific program environment.

Course prerequisites (course prerequisite quiz = practice final from course 1):

- All of the material from Course 1, “**Kalman Filter Boot Camp**,” including especially:
 - A detailed understanding of multivariable probability and stochastic processes,
 - Working knowledge of the mechanics of stochastic state-space models, and
 - Experience implementing the standard KF steps in Octave and interpreting results.